

APEX FOOD ANALYTICAL SERVICES PVT. LTD.

ISO/IEC 17025:2017 Accredited Testing Laboratory (Fictional Laboratory)

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CERTIFICATE OF ANALYSIS / LABORATORY TEST REPORT

1. GENERAL REPORT & SAMPLE IDENTIFICATION			
Report No.	Reference	AFAS/2026/HC-0914	Report Date
10-Sep-2026			
Sample Date	Received	04-Sep-2026	Analysis Started
05-Sep-2026			
Analysis Completed Date		09-Sep-2026	Mode of Receipt
			Cold chain courier box
Sample Name / Description		Raw Honey (Natural Apiary)	Batch Identification
			HC-SEP-FLOW1
Client Name		HoneyChain Traceability Project	Sampling Done By
			Client Representative
Sample Quantity Received		500 g × 2 Sealed Glass Jars	Botanical Origin
			Multifloral (Wild Forest)
Sample Condition		Intact, sealed glass containers	Storage Condition
			Ambient (22°C ± 2°C)
Testing Purpose		Honey Quality & Purity Verification	Regulatory Ref.
			FSSAI Standards 2020

Scope of Analysis: Comprehensive quality parameters, sugar spectrum profiling, physicochemical properties, and advanced adulteration screening (EA-IRMS / LC-MS/MS) performed as per standardized BIS, IS 4941, AOAC, and Codex Alimentarius analytical guidelines.

2. Organoleptic & Physical Examination

Test Parameter	Standard Reference	Observed Result	Status
Appearance & Physical State	IS 4941:2021	Viscous fluid, uniform clarity, free from debris	PASS
Color Rating (Pfund Scale)	AOAC 985.25	Amber (65 mm ± 2 mm)	PASS
Odour & Aroma Profile	Organoleptic Panel	Characteristic pleasant floral aroma, free from off-odours	PASS
Taste Profile	Organoleptic Panel	Sweet, rich floral flavor, no sour or fermented taste	PASS
Foreign Matter / Debris	IS 4941:2021	Absent	PASS
Crystallization State	Visual Observation	Liquid phase; no coarse sugar crystallization	PASS

3. Physicochemical Analysis

Test Parameter	Test Method	Observed Value	FSSAI / Codex Limit	Status
Moisture Content	IS 4941:2021	17.50 %	Max 20.0 %	PASS
Hydroxymethylfurfural (HMF)	AOAC 980.23	14.20 mg/kg	Max 40.0 mg/kg	PASS
Diastase Activity	AOAC 958.09	16.80 Schade Units	Min 8.0 Schade Units	PASS
Apparent Reducing Sugars	IS 4941:2021	71.40 %	Min 65.0 %	PASS
Sucrose Content	AOAC 977.20	1.85 %	Max 5.0 %	PASS
Fructose to Glucose Ratio	Calculation	1.18	Min 1.00	PASS
Free Acidity	IS 4941:2021	21.50 meq/kg	Max 50.0 meq/kg	PASS
Electrical Conductivity	Codex Stan 12-1981	0.48 mS/cm	Max 0.80 mS/cm	PASS
Water Insoluble Solids	IS 4941:2021	0.03 %	Max 0.10 %	PASS
Specific Optical Rotation	AOAC 925.36	−15.40°	Negative value	PASS
Ash Content	IS 4941:2021	0.18 %	Max 0.50 %	PASS
pH Value (10% solution)	Potentiometric	3.92	3.50 – 5.50	PASS

4. Advanced Purity & Adulteration Screening

Purity Test Parameter	Analytical Technique	Observed Result	Permissible Limit	Status
δ ¹³ C Protein Value	EA-IRMS	−26.50 ‰ PDB	Standard Reference	PASS
δ ¹³ C Honey Value	EA-IRMS	−26.10 ‰ PDB	Standard Reference	PASS
δ ¹³ C Difference (Δδ ¹³ C)	Calculation	0.40 ‰	≤ 1.0 ‰	PASS
C4 Sugar Adulteration	EA-IRMS	1.20 %	Max 7.0 %	PASS
Foreign Oligosaccharides	HPLC-PAD	Not Detected	Not Detected	PASS
Rice Syrup Marker (SM-R)	LC-MS/MS	Not Detected	Not Detected	PASS
Cane / Corn Syrup Markers	NMR Profiling	Not Detected	Not Detected	PASS
Commercial Sugar Screening	TLC / LC-MS	Negative	Negative	PASS

Microscopic Pollen Analysis Note: Melissopalynological examination reveals typical multi-floral honey pollen spectrum (predominantly *Brassica spp.*, *Syzygium spp.*, and wild forest species). Plant tissues, fungal spores, and starch granules are within natural physiological limits. No artificial colorants or added botanical extracts were detected.

5. Micro Contaminants & Safety Parameters

Test Parameter	Method Standard	Observed Result	Acceptable Limit	Status
Total Plate Count	IS 5402:2012	120 CFU/g	Max 1,000 CFU/g	PASS
Yeast and Mould Count	IS 5403:2012	15 CFU/g	Max 100 CFU/g	PASS
E. coli	IS 5887 (Part 1)	Absent / g	Absent / g	PASS
Salmonella spp.	IS 5887 (Part 3)	Absent / 25g	Absent / 25g	PASS
Heavy Metals (Lead as Pb)	ICP-MS	< 0.05 mg/kg	Max 0.10 mg/kg	PASS
Heavy Metals (Arsenic as As)	ICP-MS	< 0.02 mg/kg	Max 0.10 mg/kg	PASS
Antibiotic Residues (Multi-class)	LC-MS/MS	Below Detection	Below MRL	PASS

6. Quality Evaluation & Final Conclusion

FINAL STATEMENT OF CONFORMITY:

The submitted honey sample labeled as **Batch ID: HC-SEP-FLOW1** (Raw Honey) was evaluated against standard parameters mandated by the Food Safety and Standards Authority of India (FSSAI) and Codex Alimentarius Commission for natural floral honey.

Summary of Results:

- **Moisture Content:** Measured at **17.50%**, comfortably compliant with the maximum permissible limit of 20.0%.
- **Freshness Markers:** HMF (14.20 mg/kg) and Diastase activity (16.80 Schade units) indicate excellent freshness and absence of excessive thermal treatment.
- **Purity & Authenticity:** C4 sugar screening (1.20%) and Stable Carbon Isotope Ratio Analysis ($\Delta\delta^{13}\text{C} = 0.40\text{ ‰}$) confirm that the honey is free from C4 C-sugar syrups (corn/cane) and foreign oligosaccharides (rice syrup).

CONCLUSION: The tested honey sample **CONFORMS** to all prescribed quality, purity, and safety specifications and is judged **SUITABLE** for its intended human food consumption.

Analyzed By	Reviewed By	Approved Signatory
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